

Transformers and Attention

Knowledge-base document for the Domain-Specific RAG Chatbot

1. Transformer Architecture

The Transformer is a neural-network architecture designed to process sequences using attention mechanisms. Transformers became a major architecture for natural-language processing and are the foundation of many modern language models.

A transformer-based system can contain encoder components, decoder components, or both. Different model families use these components in different ways.

2. Self-Attention

Self-attention allows a model to relate different tokens within the same input sequence. For each token, the mechanism computes how strongly other tokens should contribute to its representation.

This allows the model to capture relationships between words or tokens that may be far apart in a sequence.

3. Query, Key, and Value

The attention mechanism uses three learned representations commonly called Query (Q), Key (K), and Value (V). Queries are compared with keys to produce attention scores, and those scores determine how values are combined.

Conceptually, attention can be described as finding relevant information for each token and combining that information into a context-aware representation.

4. Multi-Head Attention

Multi-head attention uses multiple attention heads. Each head can learn different relationships or patterns in the input. The outputs of the heads are combined to form a richer representation.

Using several heads allows the model to examine the sequence through multiple learned attention patterns rather than relying on a single attention operation.

5. Positional Information

Attention by itself does not inherently provide the order of tokens. Transformer systems therefore use positional information so that the model can distinguish different token positions.

Positional representations help the model reason about the sequence order when processing language.

6. Why Transformers Matter for LLMs

Transformers provide an effective mechanism for modeling relationships between tokens and can process many sequence elements in parallel during training.

Their attention-based architecture supports the large-scale language modeling systems used by many current LLMs.

7. Key Terms

Attention: a mechanism for weighting information from different parts of an input.

Self-attention: attention in which tokens attend to other tokens within the same sequence.

Multi-head attention: multiple attention operations performed in parallel using different learned projections.

Transformer: a neural-network architecture centered around attention mechanisms.